



**Elettropompe sommergibili con girante bicanale S-Flow**

*Submersible electropumps with S-Flow double channel impeller*

**DNB**

**SUBMERSIBLE PUMPS**



## Elettropompe sommergibili con girante bicanale S-Flow

### Submersible electropumps with S-Flow double channel impeller

|                     |                   |
|---------------------|-------------------|
| Potenze / Power:    | <b>0.8÷2.2 kW</b> |
| Mandate / Delivery: | <b>DN 65 - 80</b> |



#### Impieghi

La serie DNB trova impiego nel pompaggio e drenaggio di liquidi chiari e fognari. La particolare geometria delle giranti bicanali S-Flow permette il pompaggio di liquidi con presenza di corpi solidi in sospensione a fibra corta e risultano efficaci nelle seguenti applicazioni: acque di scarico, fognature e depuratori. Le versioni a 4 poli si prestano ad essere impiegate nelle condizioni di servizio continuo S1, e si distinguono per l'elevata silenziosità di funzionamento.

#### Caratteristiche

Tutti i componenti principali sono realizzati in Ghisa GG20. Due tenute meccaniche contrapposte in bagno d'olio, garantiscono il perfetto funzionamento.

#### Motori

- Motori asincroni 2-4 poli con rotore a gabbia di scoiattolo
- Protezione termica incorporata (DNB ...-2/220 e DNB 4 poli, termico T1 e T2 incorporato nel motore da collegare al quadro elettrico)
- Isolamento statore classe F (155°C)
- Grado protezione IP 68

#### Raffreddamento

Raffreddamento effettuato dal liquido nel quale la pompa è immersa.

#### Limiti di impiego

- Temperatura massima del liquido: 40°C con unità completamente sommersa
- Massima profondità di immersione: 20m
- Valori pH ammessi: 6-10
- Caratteristiche idrauliche valide per liquidi di densità <1,1 kg/dm<sup>3</sup>
- Tensioni ammesse: 230V-400V ±5%
- Frequenza ammessa: 50Hz ±2%

#### Application

The DNB series is used for pumping and draining clear liquid and sewage. The special S-Flow channels impeller permits the passage of short solid fibrous and is particularly suitable for wastewater and sewage treatment plants.

The 4 poles versions can be used in application where continue S1 service is needed and are characterized by their quiet operation.

#### Characteristic

The main components are realised in cast iron GG20. The Double mechanical seals in a back to back configuration located in the oil chamber guarantee long durability to the product.

#### Motor range

- Squirrel cage motor in 2 and 4 pole version
- Thermal protection embedded in the winding (In the DNB ...-2/220 and DNB 4 poles model the thermal protection T1 and T2 conductor to be wired to the control panel)
- Insulation class F 155°C
- Motor protection IP 68

#### Motor cooling

The cooling of the motor is ensured by the liquid where the pump is submerged.

#### Limits of use

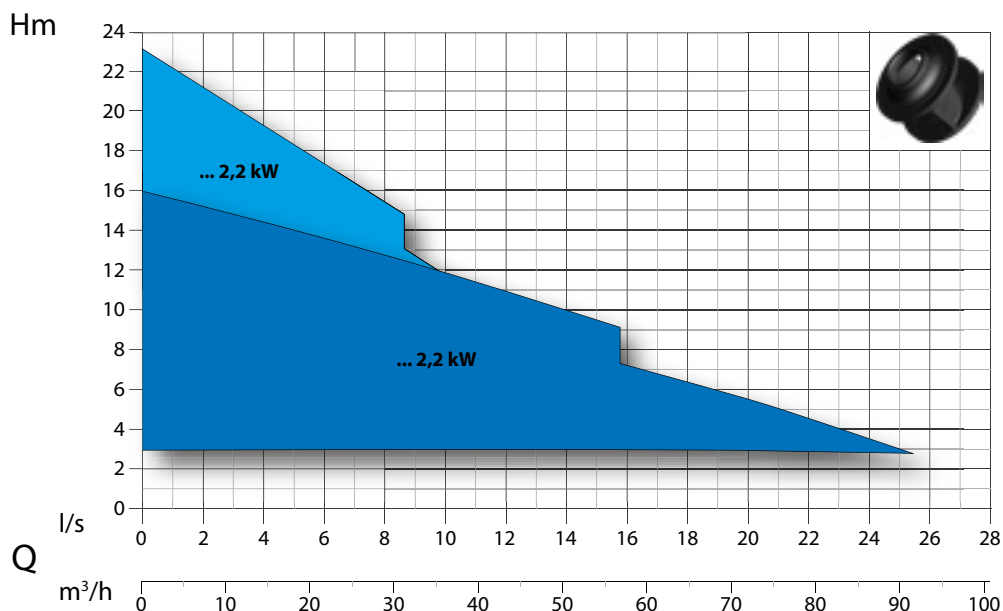
- Max. permissible liquid temperature: 40°C with pump fully submerged
- Maximum depth of immersion: 20 mt.
- Permissible pH value: 6-10
- Hydraulic features suitable for liquids with density <1,1 kg/dm<sup>3</sup>
- Allowed voltage: 230V-400V ±5%
- Allowed frequency: 50Hz ±2%

### Designazione / Designation

#### DNB EX 80 - 2 / 150 M/T G

|                                                              |  |
|--------------------------------------------------------------|--|
| Galleggiante<br>Float switch                                 |  |
| Monofase/Trifase<br>Monophase/Threephase                     |  |
| Potenza kW/P2 - es. 150=1.5kW<br>Power kW/P2 - ex. 150=1.5kW |  |
| Numero poli<br>Poles number                                  |  |
| Bocca di mandata DN<br>Delivery DN                           |  |
| Certificazione ATEX<br>ATEX certification                    |  |
| Serie pompa<br>Pump series                                   |  |

### Campo di Prestazione / Performance Overview



### Identificazione Curve

#### Curves Identification

- DN65
- DN80

### Normative

#### Norms

Curve secondo ISO 9906 livello 2  
According to ISO 9906 level 2

## Distinta dei componenti e materiali

### List of components and materials

**DNB**

Manico - Handle

Acciaio Inox - Stainless steel

Coperchio Motore - Motor Cover

Ghisa GG20 - Cast Iron GG20

Cassa Motore - Motor Casing

Ghisa GG20 - Cast Iron GG20

Albero Motore - Motor Shaft

Acciaio Inox AISI 420 - Stainless steel AISI 420

Motore elettrico - Electric Motor

Flangia Porta cuscinetto - Flange Bearing Support

Ghisa GG20 - Cast Iron GG20

Doppia Tenuta Meccanica - Double Mechanical Seal

Lato motore carbone/ceramica - Motor side: Carbon/Ceramic (CA/CE/Viton)

Lato girante carburo di silicio - Impeller side: Silicon carbide (SIC/SIC/Viton)

Configurazione ATEX - ATEX Configuration

Lato motore : tenuta a labbro - Motor side: Lip seal

Lato girante carburo di silicio - Impeller side: Silicon carbide (SIC/SIC/Viton)

Tenuta a labbro - Lips seal

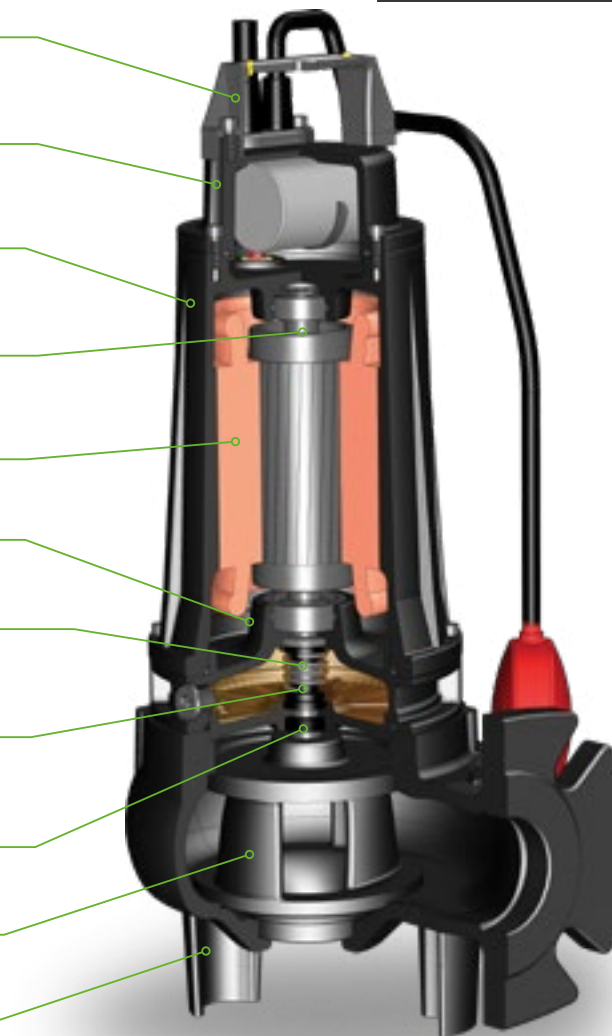
Gomma NBR - NBR rubber

Girante - Impeller

Ghisa GG20 - Cast Iron GG20

Corpo Pompa - Body Pump

Ghisa GG20 - Cast Iron GG20



## Tecnologie e Soluzioni

### Technology and Features



#### Giranti S-Flow

La superiore efficienza dei Giranti S-Flow garantisce ottime prestazioni con ampio passaggio di corpi solidi, il tutto su pompe di piccole potenze.

#### S-Flow impellers

The Superior efficiency of the S-Flow Impeller guarantees excellent performances, with wide solid bodies passage and with low power pumps.



#### Pompe antideflagranti

Tutta la serie è disponibile con certificato ATEX su richiesta

#### Explosion proof pumps

Pumps with explosion proof available on request.



0477



II 2G

Ex db IIB T4 Gb

Ex h IIB T4 Gb

0° ≤ Ta ≤ 40°

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#### Relé di Comando

Per il funzionamento del galleggiante su motori trifase fino a 1,5 kW 2 poli.

ATEX escluso.

#### Relay

For the correct operating of the float switch on the three-phase version, up to 1,5 kW 2 poles.

ATEX excluded.



#### Camera Olio ispezionabile

Totale Lubrificazione tenute garantita anche nelle condizioni più estreme. La presenza di un paraolio, assicura sempre la massima tenuta in caso di traflaggio d'olio.

#### Inspectional Oil Chamber

Excellent lubrication of the mechanical seals even in the harshest pumping conditions. The presence of a lip seal ensure the maximum seal in case of oil leakage.

### Mandata Orizzontale DN65 PN10 - RPM 2850 1/min 2 poli

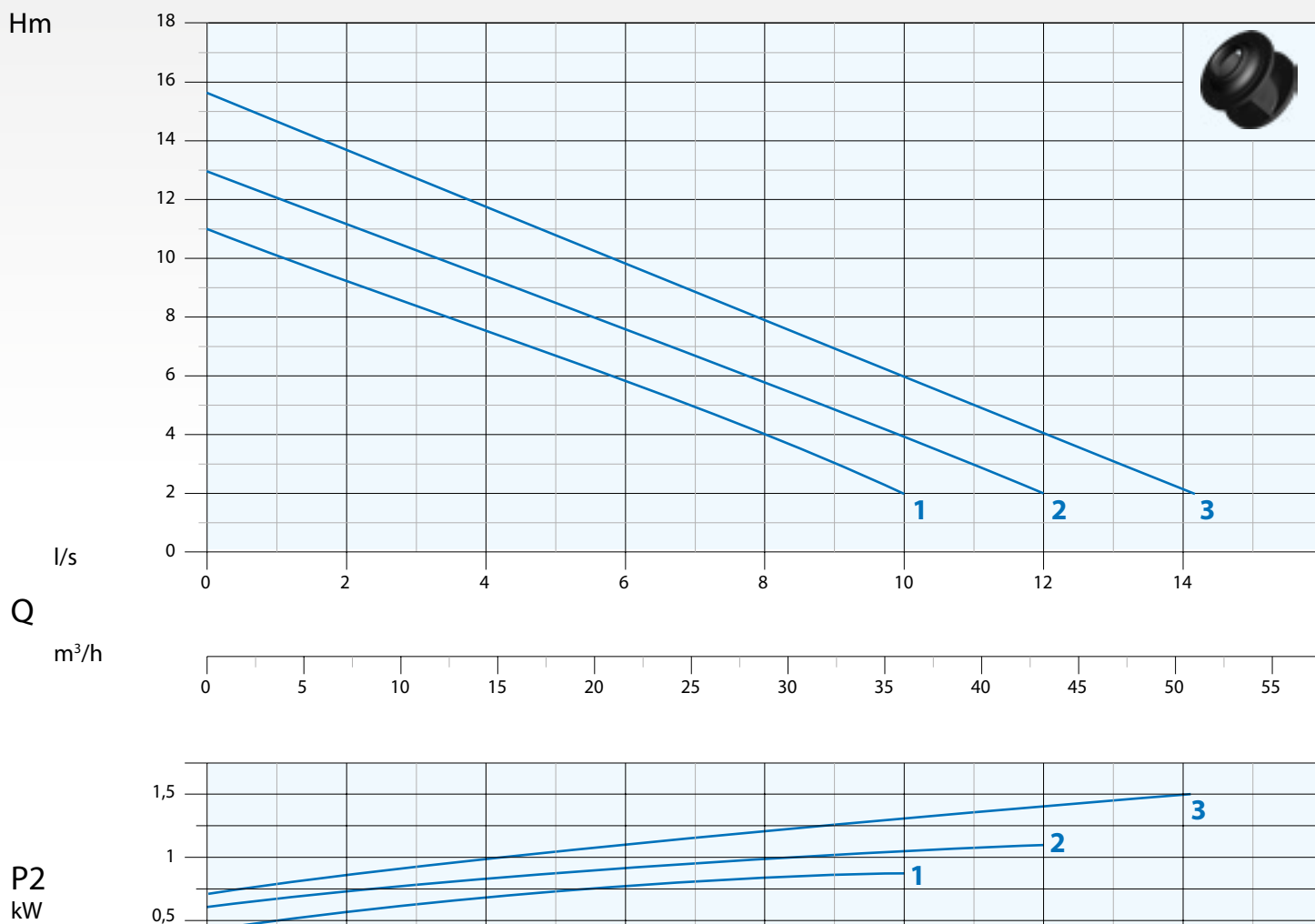
Horizontal Outlet DN65 PN10 - RPM 2850 1/min 2 poles

Immagine a solo scopo illustrativo  
Picture for illustration purposes only



### Curva di Prestazione

Performance Curve



| N° | Tipo Type        |      |      |      |      |      |      |      |      |      |      |
|----|------------------|------|------|------|------|------|------|------|------|------|------|
|    |                  | I/s  | 1    | 2    | 4    | 6    | 8    | 10   | 11   | 12   | 14   |
|    |                  | I/m  | 60   | 120  | 240  | 360  | 480  | 600  | 660  | 720  | 840  |
|    |                  | m³/h | 3,6  | 7,2  | 14,4 | 21,6 | 28,8 | 36,0 | 39,6 | 43,2 | 50,4 |
| 1  | DNB 65-2/080 M/T | mt   | 10   | 9    | 7,5  | 5,5  | 3,5  | 2    |      |      |      |
| 2  | DNB 65-2/110 M/T |      | 12,5 | 11,5 | 9,5  | 7,5  | 5,5  | 4    | 3    |      |      |
| 3  | DNB 65-2/150 M/T |      | 15   | 14   | 12   | 10   | 8    | 6    | 5    | 4    | 2    |

| N° | Tipo Type        | EX | Mandata Delivery | Passaggio Free Passage | kW  |     | HP  | R.P.M. 1/min | A              |    |                | Hz |
|----|------------------|----|------------------|------------------------|-----|-----|-----|--------------|----------------|----|----------------|----|
|    |                  |    |                  |                        | P1  | P2  |     |              | 1 Phase - 230V | µf | 3 Phase - 400V |    |
| 1  | DNB 65-2/080 M/T | •  | DN65 PN10 - G 2" | 50 mm                  | 1,1 | 0,8 | 1,0 | 2850         | 5,2            | 18 | 2,1            | 50 |
| 2  | DNB 65-2/110 M/T | •  |                  |                        | 1,5 | 1,1 | 1,5 |              | 7,5            | 30 | 2,8            |    |
| 3  | DNB 65-2/150 M/T | •  | DN65 PN10        |                        | 2,0 | 1,5 | 2   |              | 9,9            | 32 | 3,7            |    |

• Pompa antideflagrante disponibile con certificazione  
Available explosion proof pump

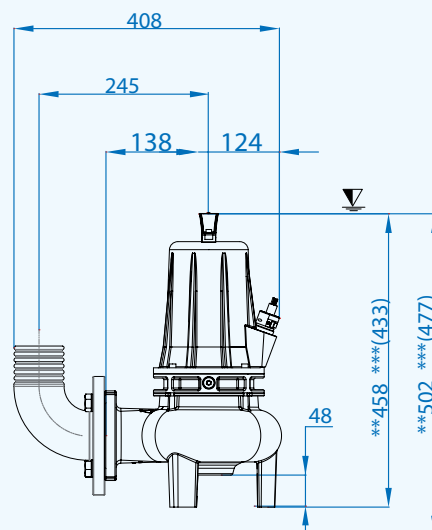
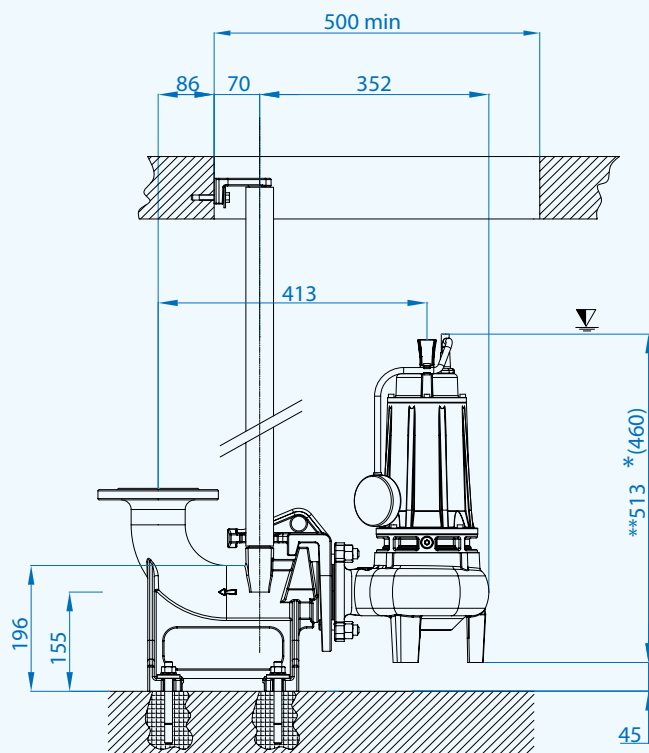


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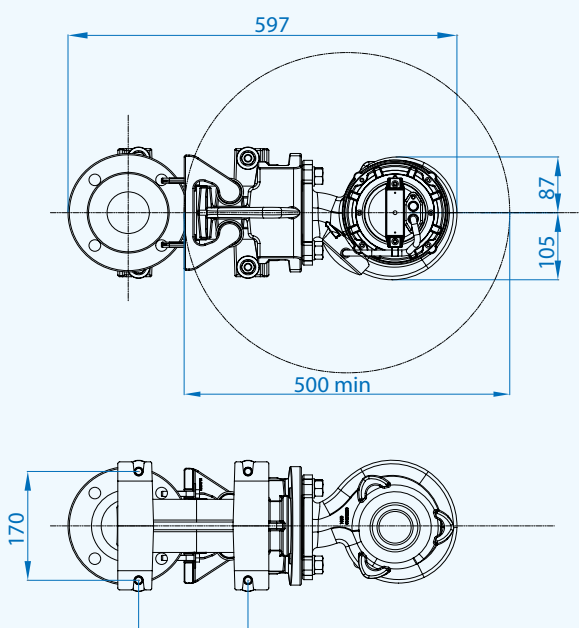


II 2G Ex db IIB T4 Gb  
Ex h IIB T4 Gb  
0° ≤ Ta ≤ 40°

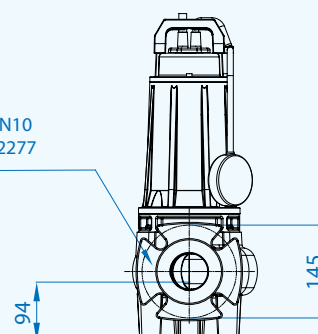
▽ Sommergenza minima  
Minimum submersion



\*(DNB 65-2/080)  
\*\*(DNB 65-2/110-150)  
\*\*\* (VERSIONI ATEX - ATEX VERSIONS)



DN65 PN10  
ex UNI 2277



## Cavi / Cables

| Versione<br>Version       | Fasi<br>Phases | Cavo<br>Cable | Sezione cavo mm <sup>2</sup><br>Cable cross section mm <sup>2</sup> | mt |
|---------------------------|----------------|---------------|---------------------------------------------------------------------|----|
| Standard                  | 1 ~ 230V       | H07RN8F       | 3x1 Ø9*                                                             | 10 |
|                           | 3 ~ 400V       |               | 4x1 Ø10***                                                          | 10 |
| Versioni/Versions<br>ATEX | 1 ~ 230V       | NSSHÖU-J      | 4x1,5+2x0,50 Ø12**                                                  | 10 |
|                           | 3 ~ 400V       |               | 4x1,5+2x0,50 Ø12***                                                 | 10 |

\* Spina Schuko - Schuko plug

\*\* Quadro di comando - Control box

\*\*\* Terminali liberi - Free terminals

## Dimensioni imballo / Packaging dimensions

| Tipo - Type      | X<br>mm | Y<br>mm | Z<br>mm | Kg |
|------------------|---------|---------|---------|----|
| DNB 65-2/080 M/T | 206     | 585     | 313     | 21 |
| DNB 65-2/110 M/T |         |         |         | 29 |
| DNB 65-2/150 M/T |         |         |         | 31 |

## Accessori - Optional



Quadri ATEX disponibili su richiesta  
Explosion proof control box available on request



Pompe monofasi:  
Control-box per  
funzionamento con  
condensatore esterno  
For single phase pumps:  
Control-box with external  
main capacitor



Piede di accoppiamento  
rapido Tipo:  
DUTY 50 e DUTY 65 e  
B4(PN10)  
Automatic coupling foot  
Type: DUTY 50 - DUTY 65 - B4  
(PN10)



Curva flangiata N5 e  
Curva filettata G2" con  
portagomma fino a 1,1 kW  
Flanged hose connection N5  
Thread G2" hose connection  
up to 1,1 kW

Doc\_Rev.1

Date\_01/02/18

### Mandata Orizzontale DN65 PN10 - RPM 2850 1/min 2 poli

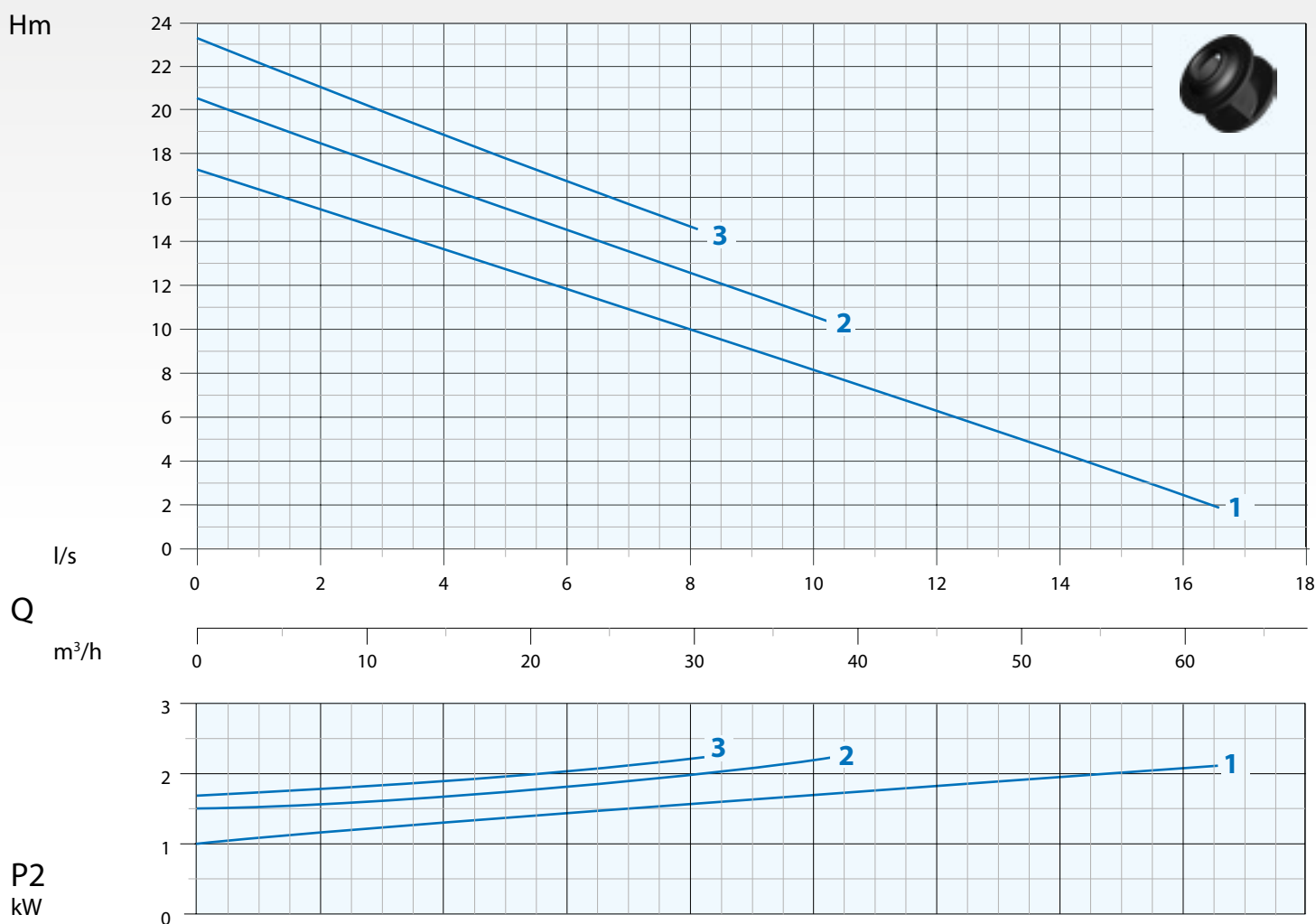
Horizontal Outlet DN65 PN10 - RPM 2850 1/min 2 poles

Immagine a solo scopo illustrativo  
Picture for illustration purposes only



### Curva di Prestazione

Performance Curve



| N° | Tipo<br>Type     | l/s  | 1    | 2    | 3    | 4    | 6    | 8    | 10   | 12   | 14   | 16   |
|----|------------------|------|------|------|------|------|------|------|------|------|------|------|
|    |                  | l/m  | 60   | 120  | 180  | 240  | 360  | 480  | 600  | 720  | 840  | 960  |
|    |                  | m³/h | 3,6  | 7,2  | 10,8 | 14,4 | 21,6 | 28,8 | 36   | 43,2 | 50,4 | 57,6 |
| 1  | DNB 65-2/220 T   |      | 17,5 | 16,5 | 15,5 | 14,5 | 12,5 | 10,5 | 8,5  | 6,5  | 4,5  | 2,5  |
| 2  | DNB 65-2/220-1 T |      | 20   | 19   | 18   | 17   | 14,5 | 12,5 | 10,5 |      |      |      |
| 3  | DNB 65-2/220-2 T |      | 22,5 | 21   | 20   | 19   | 17   | 14,5 |      |      |      |      |

| N° | Tipo<br>Type     | EX | Mandata<br>Delivery | Passaggio<br>Free Passage | kW  |     | HP | R.P.M.<br>1/min | 3 Phase - 400V | Hz |
|----|------------------|----|---------------------|---------------------------|-----|-----|----|-----------------|----------------|----|
|    |                  |    |                     |                           | P1  | P2  |    |                 |                |    |
| 1  | DNB 65-2/220 T   | •  | DN65 PN10           | 50 mm                     | 2,5 | 2,2 | 3  | 2850            | 5              | 50 |
| 2  | DNB 65-2/220-1 T | •  |                     |                           | 2,9 | 2,2 | 3  |                 | 5,3            |    |
| 3  | DNB 65-2/220-2 T | •  |                     |                           | 3   | 2,2 | 3  |                 | 5,3            |    |

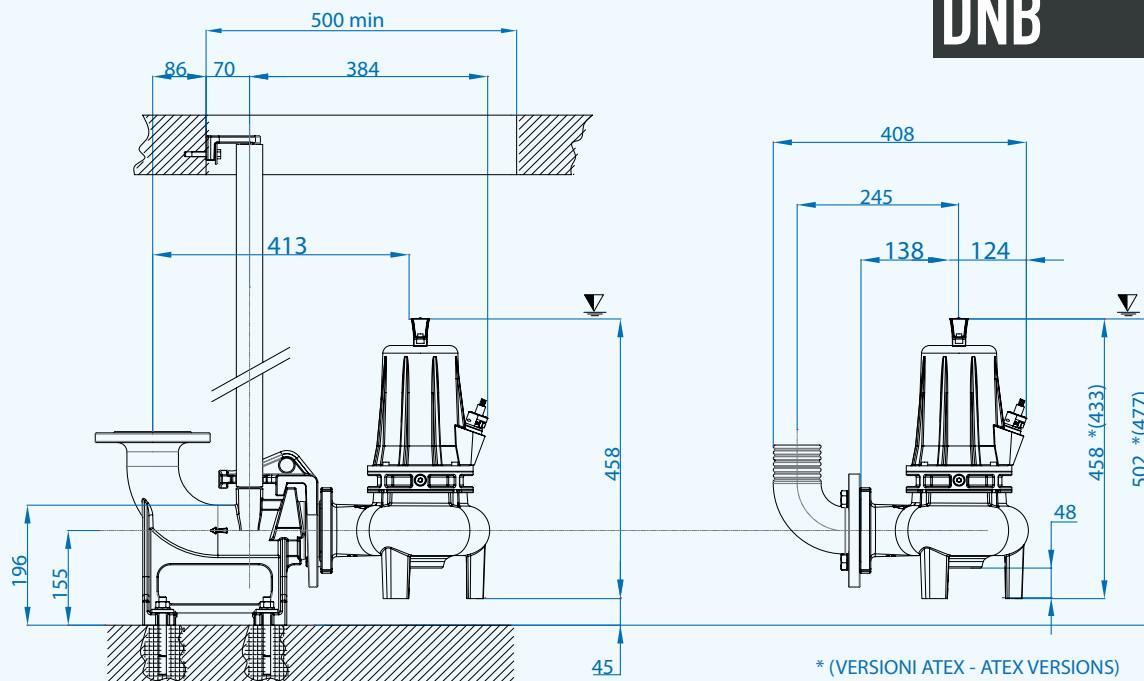
• Pompa antideflagrante disponibile con certificazione  
Available explosion proof pump



Ex db IIB T4 Gb  
Ex h IIB T4 Gb  
0° ≤ Ta ≤ 40°

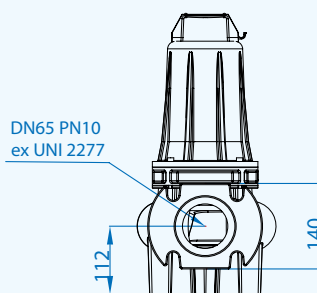
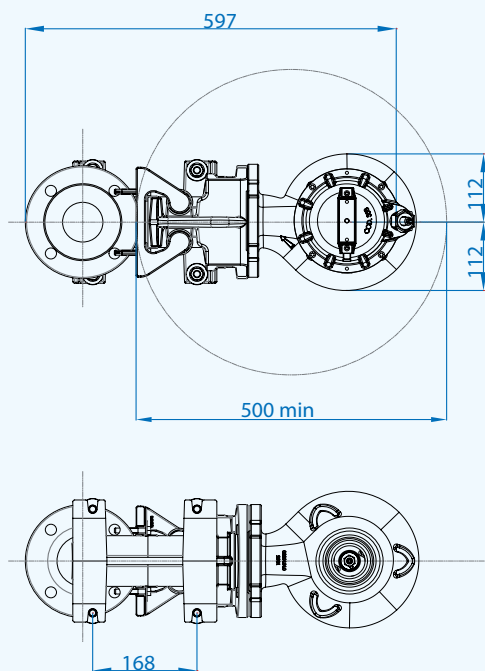
EPT 17 ATEX 2701 X





\*(VERSIONI ATEX - ATEX VERSIONS)

▼ Sommergenza minima  
Minimum submersion



## Cavi / Cables

| Versione<br>Version       | Fasi<br>Phases | Cavo<br>Cable | Sezione cavo mm <sup>2</sup><br>Cable cross section mm <sup>2</sup> | mt |
|---------------------------|----------------|---------------|---------------------------------------------------------------------|----|
| Standard                  | 3 ~ 400V       | H07RN8F       | 4x1,5+2x0,50 Ø12*                                                   | 10 |
| Versioni/Versions<br>ATEX | 3 ~ 400V       | NSSHÖU-J      | 4x1,5+2x0,50 Ø14*                                                   | 10 |

\* Terminali liberi - Free terminals

## Dimensioni imballo / Packaging dimensions

| Tipo - Type      | X<br>mm | Y<br>mm | Z<br>mm | Kg |
|------------------|---------|---------|---------|----|
| DNB 65-2/220 T   |         |         |         | 35 |
| DNB 65-2/220-1 T | 206     | 585     | 313     | 36 |
| DNB 65-2/220-2 T |         |         |         | 36 |

## Accessori - Optional



Quadri ATEX disponibili su richiesta  
Explosion proof control box available on request



Pompe monofasi:  
Control-box per  
funzionamento con  
condensatore esterno  
For single phase pumps:  
Control-box with external  
main capacitor



Piede di accoppiamento  
rapido Tipo:  
DUTY 65 e B4/PN10  
Automatic coupling foot  
Type: DUTY 65 -B4/PN10



Curva flangiata N5  
Flanged hose connection N5

Doc\_Rev.1  
Date\_01/02/18



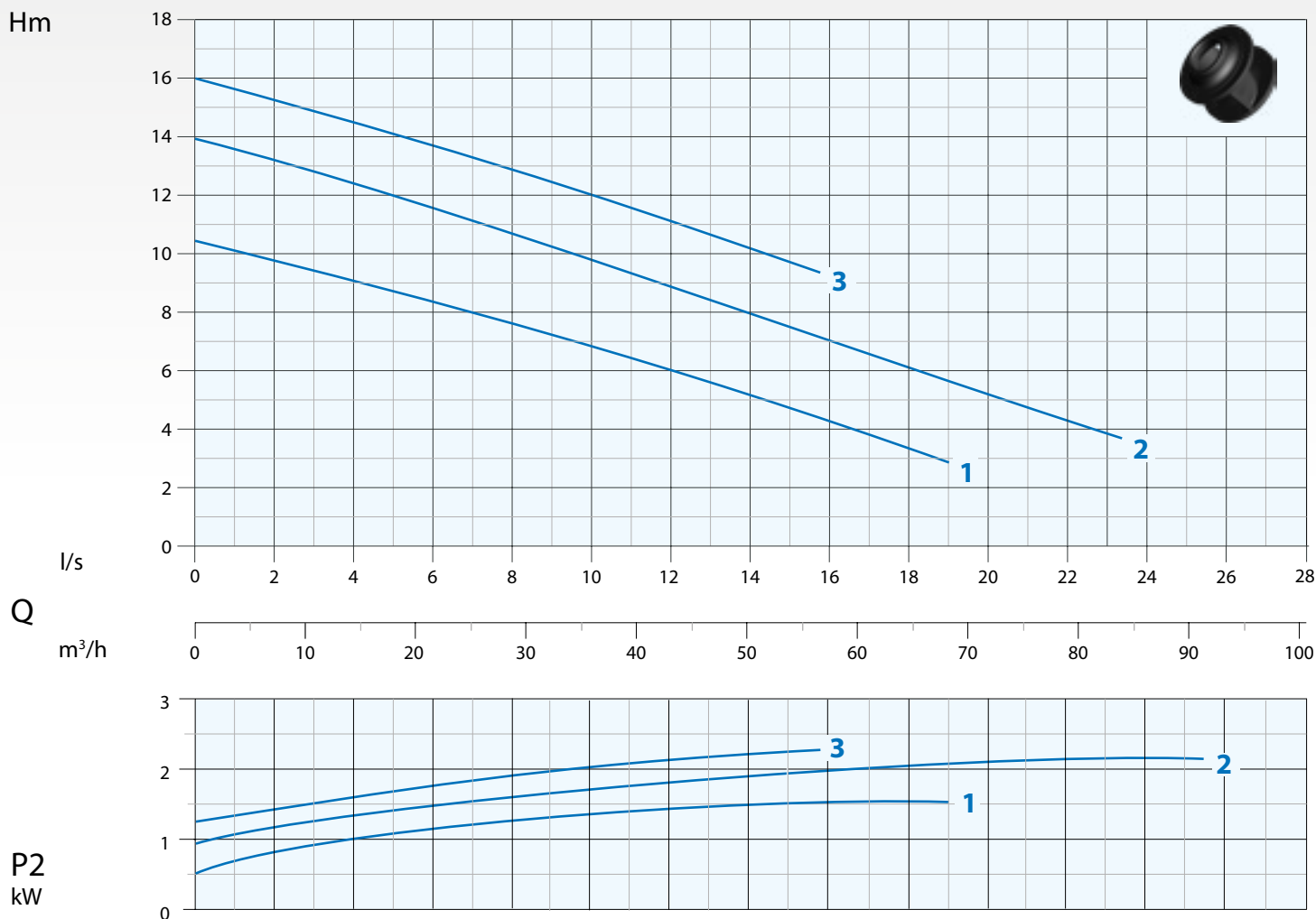
### Mandata Orizzontale DN80 PN16 - RPM 2850 1/min 2 poli

Horizontal Outlet DN80 PN16 - RPM 2850 1/min 2 poles

### Curva di Prestazione

Performance Curve

Immagine a solo scopo illustrativo  
Picture for illustration purposes only



| N° | Tipo Type        | l/s  | 2   | 4    | 6    | 8    | 10  | 12   | 14   | 16   | 18   | 20   | 22   | 24   | 25   |
|----|------------------|------|-----|------|------|------|-----|------|------|------|------|------|------|------|------|
|    |                  | l/m  | 120 | 240  | 360  | 480  | 600 | 720  | 840  | 960  | 1080 | 1200 | 1320 | 1440 | 1500 |
|    |                  | m³/h | 7,2 | 14,4 | 21,6 | 28,8 | 36  | 43,2 | 50,4 | 57,6 | 64,8 | 72   | 79,2 | 86,4 | 90   |
| 1  | DNB 80-2/150 M/T | mt   | 9,5 | 9    | 8,5  | 7,5  | 7   | 6    | 5    | 4,5  | 3,5  |      |      |      |      |
| 2  | DNB 80-2/220 T   |      | 13  | 12,5 | 11,5 | 10,5 | 9,5 | 9    | 8    | 7    | 6,5  | 5,5  | 4,5  | 3,5  | 3    |
| 3  | DNB 80-2/220-1 T |      | 15  | 14,5 | 13,5 | 13   | 12  | 11   | 10   | 9,5  |      |      |      |      |      |

| N° | Tipo Type        | EX | Mandata Delivery | Passaggio Free Passage | kW  |     | HP | R.P.M. 1/min | A              |    |                | Hz |
|----|------------------|----|------------------|------------------------|-----|-----|----|--------------|----------------|----|----------------|----|
|    |                  |    |                  |                        | P1  | P2  |    |              | 1 Phase - 230V | μf | 3 Phase - 400V |    |
| 2  | DNB 80-2/150 M/T | •  | DN80 PN16        | 70 mm                  | 2,1 | 1,5 | 2  | 2850         | 9,9            | 32 | 3,7            | 50 |
| 3  | DNB 80-2/220 T   | •  |                  |                        | 2,9 | 2,2 | 3  |              | -              | -  | 5,5            |    |
| 4  | DNB 80-2/220-1 T | •  |                  |                        | 2,9 | 2,2 | 3  |              | -              | -  | 5,3            |    |

• Pompa antideflagrante disponibile con certificazione  
Available explosion proof pump

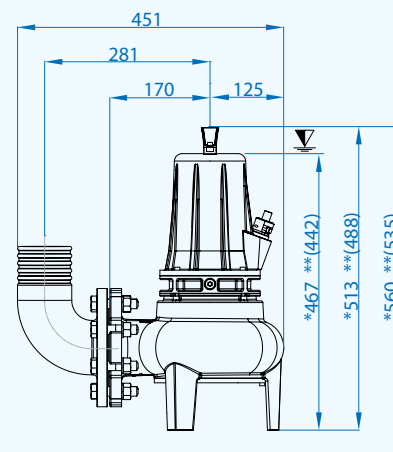
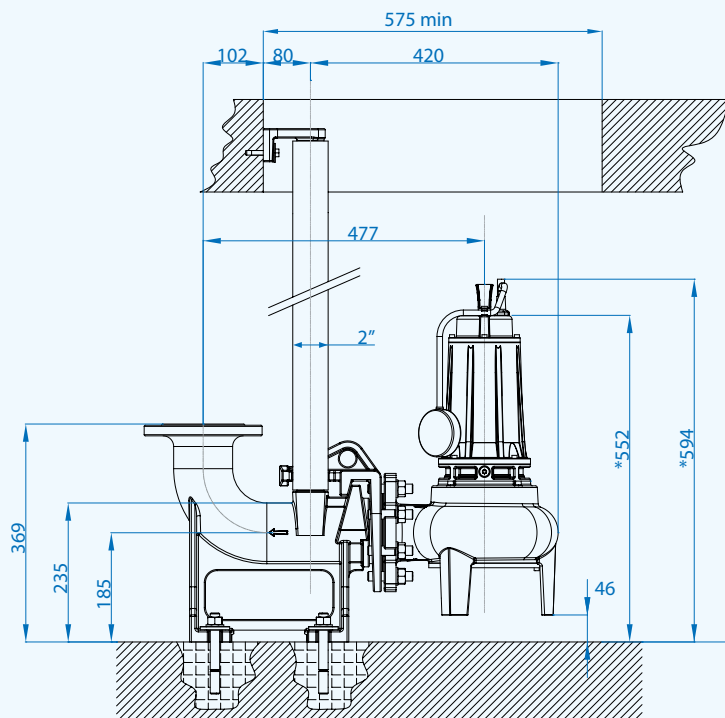


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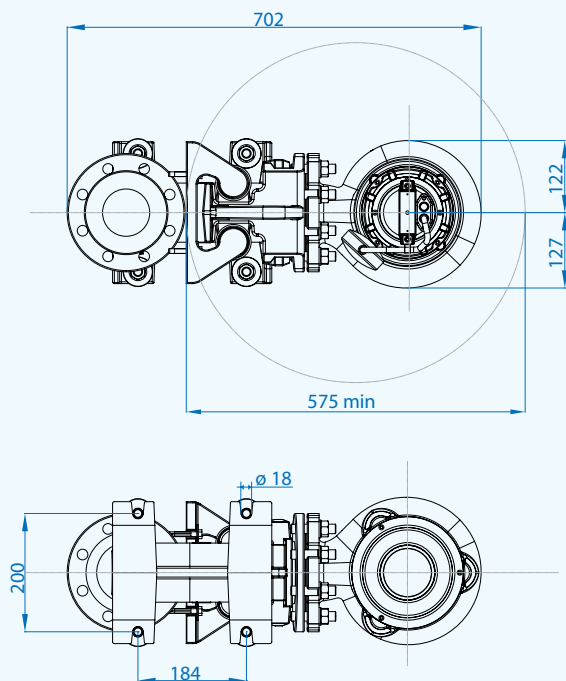


Ex db IIB T4 Gb  
Ex h IIB T4 Gb  
0° ≤ Ta ≤ 40°

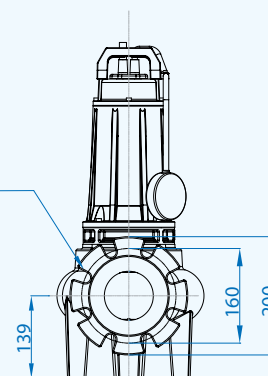
▼ Sommergenza minima  
Minimum submersion



\*DNB 80-2/150  
\*\*(DNB 80-2/220 e VERSIONI ATEX  
DNB 80-2/220 and ATEX VERSION)



DN80 PN16  
ex UNI 2278



## Cavi / Cables

| Pompe<br>Pumps            | Fasi<br>Phases | Cavo<br>Cable | Sezione cavo mm <sup>2</sup><br>Cable cross section mm <sup>2</sup> | mt |
|---------------------------|----------------|---------------|---------------------------------------------------------------------|----|
| DNB 80-2/150              | 1 ~ 230V       | H07RN8F       | 3x1 Ø9*                                                             | 10 |
|                           | 3 ~ 400V       |               | 4x1 Ø10***                                                          | 10 |
| DNB 80-2/220 (-1)         | 3 ~ 400V       | H07RN8F       | 4x1,5+2x0,50 Ø12***                                                 | 10 |
| Versioni/Versions<br>ATEX | 1 ~ 230V       | NSSHÖU-J      | 4x1,5+2x0,50 Ø12**                                                  | 10 |
|                           | 3 ~ 400V       |               | 4x1,5+2x0,50 Ø12***                                                 | 10 |

\* Spina Schuko - Schuko plug

\*\* Quadro di comando - Control box

\*\*\* Terminali liberi - Free terminals

## Dimensioni imballo / Packaging dimensions

| Tipo - Type      | X<br>mm | Y<br>mm | Z<br>mm | Kg |
|------------------|---------|---------|---------|----|
| DNB 80-2/150 M/T | 206     | 585     | 313     | 35 |
| DNB 80-2/220 T   |         |         |         | 36 |
| DNB 80-2/220-1 T |         |         |         | 35 |

## Accessori - Optional



Quadri ATEX disponibili su richiesta  
Explosion proof control box available  
on request



Pompe monofasi:  
Control-box per  
funzionamento con  
condensatore esterno  
For single phase pumps:  
Control-box with external  
main capacitor



Piede di accoppiamento  
rapido Tipo: DUTY 80 e B5  
Automatic coupling foot  
Type: DUTY 80 and B5



Curva flangiata  
portagomma N2  
Flanged hose connection N2

Doc\_Rev.1

Date\_01/02/18

## Mandata Orizzontale DN80 PN16 - RPM 1450 1/min 4 poli

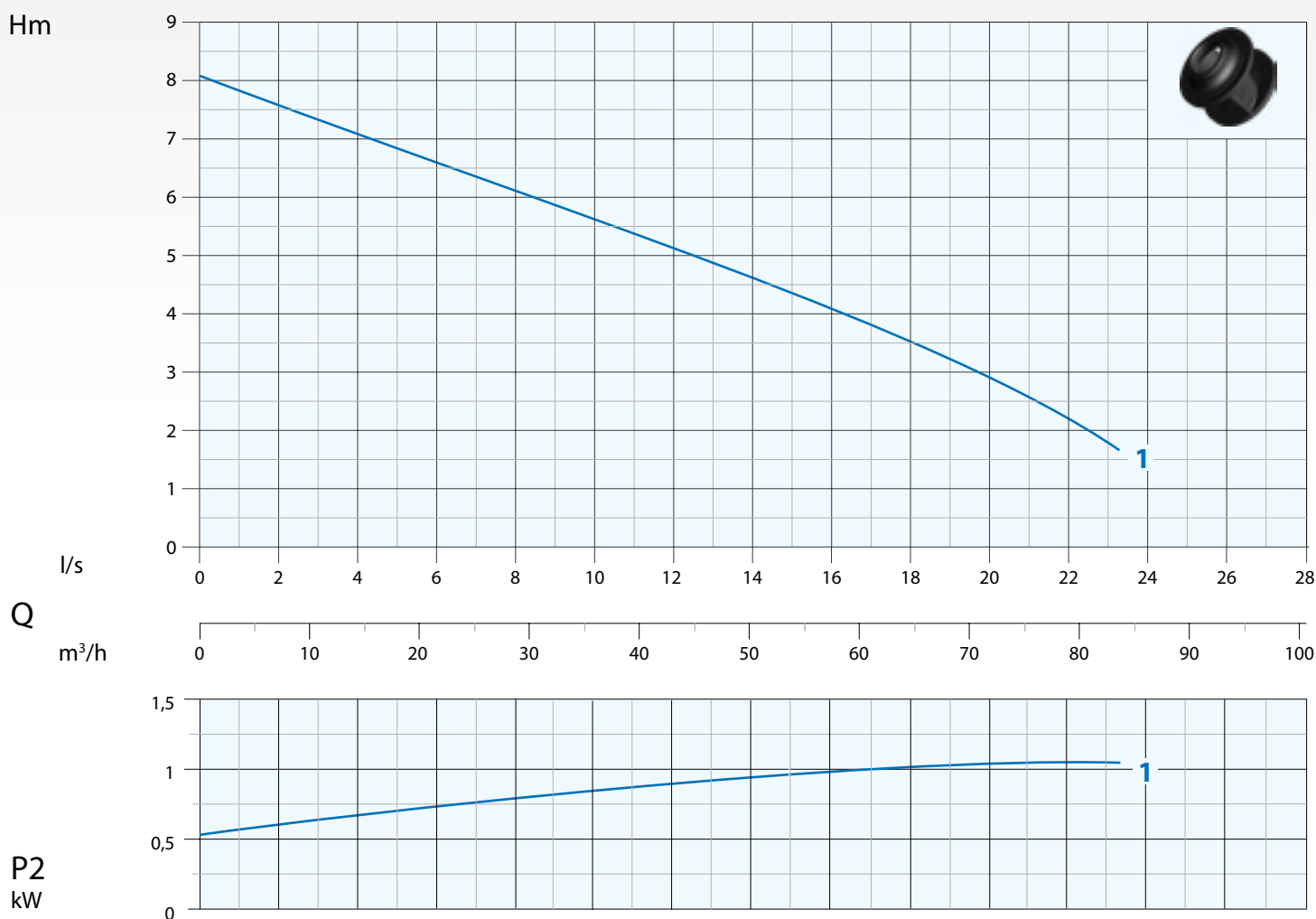
Horizontal Outlet DN80 PN16 - RPM 1450 1/min 4 poles

Immagine a solo scopo illustrativo  
Picture for illustration purposes only



## Curva di Prestazione

Performance Curve



| N° | Tipo Type | I/s  | 1   | 2   | 4    | 6    | 8    | 10  | 12   | 14   | 16   | 18   | 20   | 22   |
|----|-----------|------|-----|-----|------|------|------|-----|------|------|------|------|------|------|
|    |           | I/m  | 60  | 120 | 240  | 360  | 480  | 600 | 720  | 840  | 960  | 1080 | 1200 | 1320 |
|    |           | m³/h | 3,6 | 7,2 | 14,4 | 21,6 | 28,8 | 36  | 43,2 | 50,4 | 57,6 | 64,8 | 72   | 79,2 |
|    |           | mt   | 7,8 | 7,5 | 7    | 6,5  | 6    | 5,5 | 5    | 4,5  | 4    | 3,5  | 2,8  | 2,5  |

| N° | Tipo Type        | EX | Mandata Delivery | Passaggio Free Passage | kW  |     | HP  | R.P.M. 1/min | A              |    |                | Hz |
|----|------------------|----|------------------|------------------------|-----|-----|-----|--------------|----------------|----|----------------|----|
|    |                  |    |                  |                        | P1  | P2  |     |              | 1 Phase - 230V | µf | 3 Phase - 400V |    |
| 1  | DNB 80-4/110 M/T | •  | DN80 PN16        | 70 mm                  | 1,5 | 1,1 | 1,5 | 1450         | 7,6            | 35 | 3,2            | 50 |

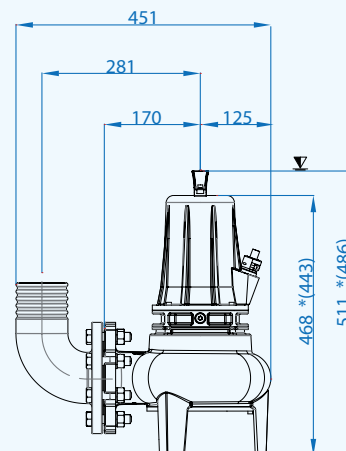
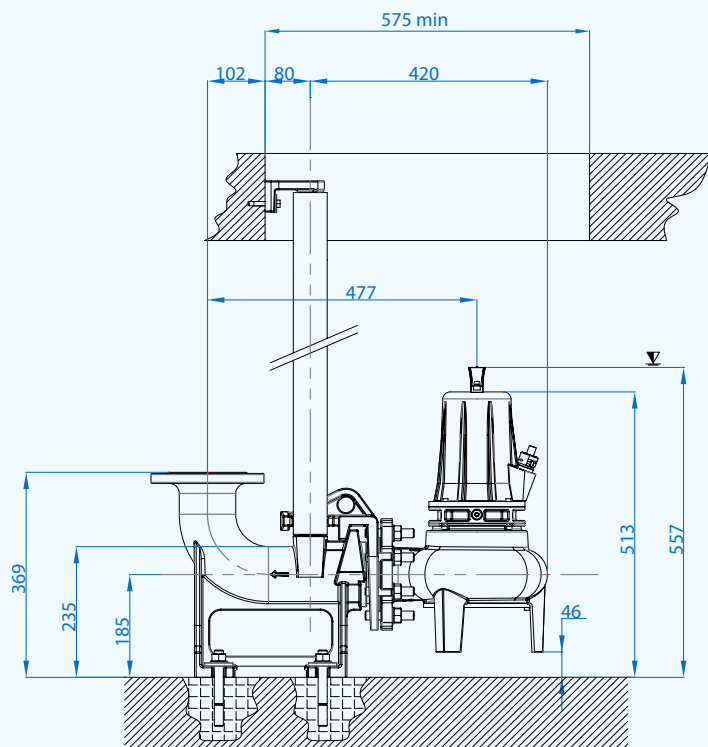
• Pompa antideflagrante disponibile con certificazione  
Available explosion proof pump



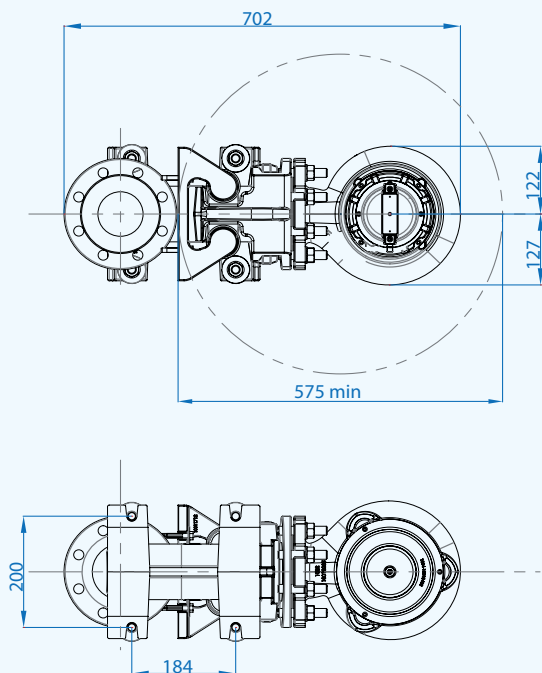
II 2G  
Ex db IIB T4 Gb  
Ex h IIB T4 Gb  
0° ≤ Ta ≤ 40°

EPT 17 ATEX 2701 X

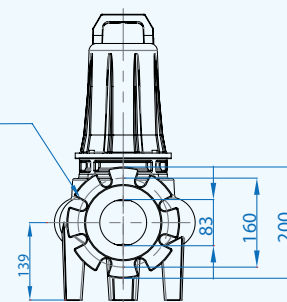
\* VERSIONI ATEX - ATEX VERSION



∇ Sommergenza minima  
Minimum submersion



DN80 PN16  
ex UNI 2278



## Cavi / Cables

|                           | Fasi<br>Phases | Cavo<br>Cable | Sezione cavo mm <sup>2</sup><br>Cable cross section mm <sup>2</sup> | mt |
|---------------------------|----------------|---------------|---------------------------------------------------------------------|----|
| Standard                  | 1 ~ 230V       | H07RN8F       | 4x1,5+2x0,50 Ø12*                                                   | 10 |
|                           | 3 ~ 400V       |               | 4x1,5+2x0,50 Ø12**                                                  | 10 |
| Versioni/Versions<br>ATEX | 1 ~ 230V       | NSSHÖU-J      | 4x1,5+2x0,50 Ø12*                                                   | 10 |
|                           | 3 ~ 400V       |               | 4x1,5+2x0,50 Ø12**                                                  | 10 |

\* Di serie con Control Box - Standard with Control Box

\*\* Terminali liberi - Free terminals

## Dimensioni imballo / Packaging dimensions

| Tipo - Type      | X<br>mm | Y<br>mm | Z<br>mm | Kg |
|------------------|---------|---------|---------|----|
| DNB 80-4/110 M/T | 260     | 585     | 315     | 36 |



## Accessori - Optional



Quadri ATEX disponibili su richiesta  
Explosion proof control box available on request



Pompe monofasi:  
Control-box per  
funzionamento con  
condensatore esterno  
For single phase pumps:  
Control-box with external  
main capacitor



Piede di accoppiamento  
rapido Tipo: DUTY 80 e B5  
Automatic coupling foot  
Type: DUTY 80 and B5



Curva flangiata  
portagomma N2  
Flanged hose connection N2

Doc\_Rev.1

Date\_01/02/18